3	**a**	The internal energy is the sum of the **random distribution of** kinetic and potential energies of all the molecules of the gas. Marking points: • sum of/total potential energy and kinetic energy of (all) molecules/particles • reference to random (distribution)	**M1** **A1**
	b(i)	Work done by system/gas = $p\Delta V$ $= 1.03 \times 10^5 \times (2.96 \times 10^{-2} - 1.87 \times 10^{-5})$ $= 3046.9$ $= 3050$ J Work done on system, W = **–** 3050 J (negative value expected)	**C1** **A1**
	b(ii)	$Q = 4.05 \times 10^4$ J	**B1**
	b(ii)	$\Delta U = Q + W$ $= 4.05 \times 10^4 - 3047$ $= 37\,500$ J internal energy increase	**A1** **B1**

	c	number of molecules = N_A energy = $37500 / (6.02 \times 10^{23})$ $= 6.23 \times 10^{-20}$ J	 **C1** **A1**
		Total:	**9**